

SnapTarget

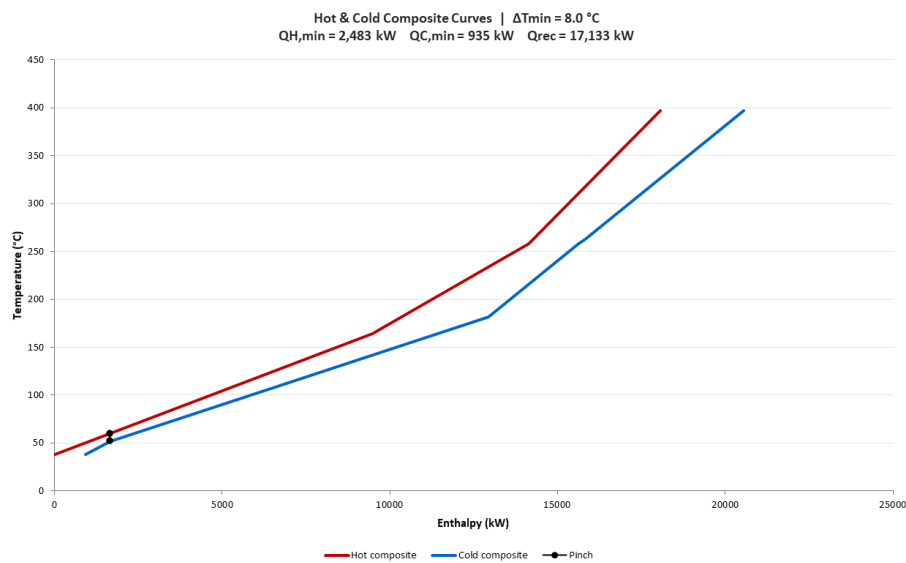
HEN Supertargeting Add-in for Microsoft Excel — User Guide

User Guide

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Created by Mohammed Ali Mansoury

Heat Exchanger Network synthesis for a Kerosene Hydrotreater — Graduation Project



SnapTarget performs pinch-analysis supertargeting inside Excel: energy, area and units targets (Problem Table Algorithm + Bath formula), the ΔT_{min} trade-off study with two capital-cost models, and seven live charts (one per sheet). It is a pure-VBA port of the validated TOP v9 model and reproduces it to machine precision (worst relative deviation 4×10^{-15} over the full sweep).

1. Installation

Try it once: double-click SnapTarget .xlam and enable macros when prompted. Install permanently: Excel → File → Options → Add-ins → Manage: Excel Add-ins → Go... → Browse... → select the file. If Windows blocks it, right-click the file → Properties → Unblock.

Workbooks created with SnapTarget use its worksheet functions, so they display #NAME? on machines where the add-in is not installed.

2. Workflow

Step	Ribbon button	What you get
1	New Template	Streams input workbook, pre-filled with the kerosene hydrotreater demo case (streams CMIX, C3, C4, H1–H3 + economics), plus Credits and License (MIT) sheets.
2	—	Set ΔT_{min} (B4) and No. of streams (E4) — the stream table instantly resizes to that many rows — then enter the streams and adjust the four input boxes.
3	Compute Targets	PTA sheet (full heat cascade, pinch highlighted) and the live Results sheet — three sections: Pinch & Energy Targets, Area & Units Targets, Cost Targets.
4	Super Targets	Supertargeting Trade-Off Study sheet (ΔT_{min} 0.5–40 °C, optima highlighted).
5	Charts	Seven live charts, one per sheet (auto-builds the trade-off study if missing).

Everything except the PTA sheet is live. The Results table, the Trade-Off Study, the chart curves and the chart captions are formulas: the moment you change a stream, ΔT_{min} or an economic parameter, they all recalculate — no button re-run needed. The PTA cascade is a snapshot (its row structure depends on the stream count); re-run Compute Targets to refresh it.

3. The Streams input sheet

SnapTarget — HEN Supertargeting
Pinch targets, trade-off and PTA from the TOP v9 model | v2.5
Set ΔT_{min} and No. of streams (the stream table resizes itself), then enter the streams; leave Type blank to auto-detect hot/cold from the temperatures. Results, the Trade-Off Study and all charts update live; re-run Compute Targets to refresh the PTA snapshot.

ΔT_{min} (°C) No. of streams

Stream	Type	Ts (°C)	Tt (°C)	CP (kW/°C)	h (kW/m ² ·°C)	Duty (kW)
CMIX	cold	52.013	397	35.2817	1.066	12,171.7
C3	cold	38	182	51.4877	0.58	7,414.2
C4	cold	258	263	6.0212	0.58	30.1
H1	hot	397	38	28.2364	0.58	10,136.9
H2	hot	258	38	21.3588	0.58	4,698.9
H3	hot	164	38	25.6489	0.65	3,231.8

Buttons: Compute Targets, Super Targets, Charts

Economics

Hot utility cost (\$/kW-yr)	128.5
Cold utility cost (\$/kW-yr)	6.426
Rate of return (-)	0.1
Plant life (yr)	10

Area & Capital Cost

Fixed cost per unit (\$)	16506
Area cost coefficient (\$/m ² ·exp)	145
Area cost exponent (-)	0.65

Utility Temperatures

Hot utility Tin (°C)	800
Hot utility Tout (°C)	350
Cold utility Tin (°C)	20
Cold utility Tout (°C)	35

Utility Film Coefficients

Hot utility h (kW/m ² ·°C)	0.111
Cold utility h (kW/m ² ·°C)	3.75

The input template: ΔT_{min} and No. of streams boxes, the self-resizing stream table (left), the four input boxes (right), action buttons.

Input	Location	Meaning
ΔT_{min}	B4	Minimum approach temperature for targeting (°C).
No. of streams	E4	Stream count (1–40). Changing it instantly resizes the stream table to that many rows; rows dropped when shrinking are cleared.
Stream rows	A7 down	Name, Type (hot/cold dropdown — or leave blank to auto-detect from the temperatures), Ts, Tt (°C), CP (kW/°C), film coefficient h (kW/m ² ·°C). The gray Duty column shows each stream's heat load Ts–Tt ·CP live.
Economics	J7:J10	Hot/cold utility cost (\$/kW-yr), rate of return, plant life.
Area & Capital Cost	J13:J15	Fixed cost per unit (\$), area cost coefficient (\$/m ² · ^{exp}) and area cost exponent.
Utility Temperatures	J18:J21	Hot utility Tin/Tout and cold utility Tin/Tout (°C).

Input	Location	Meaning
Utility Film Coefficients	J24:J25	Hot and cold utility h (kW/m ² .°C), for heater/cooler area targets.

4. Output sheets

4.1 PTA — Problem Table Algorithm

Problem Table Algorithm

$\Delta T_{min} = 8.0\text{ }^{\circ}\text{C}$ | shifted temperature scale (hot $-\Delta T/2$, cold $+\Delta T/2$) | snapshot: re-run Compute Targets after changing inputs

Stream	Type	T*hi (°C)	T*lo (°C)	CP (kW/°C)
CMIX	cold	401.0	56.0	35.2817
C3	cold	186.0	42.0	51.4877
C4	cold	267.0	262.0	6.0212
H1	hot	393.0	34.0	28.2364
H2	hot	254.0	34.0	21.3588
H3	hot	160.0	34.0	25.6489

Heat cascade

T* (°C)	ΔT (°C)	$\Sigma CPh-\Sigma CPc$ (kW/°C)	ΔH (kW)	Infeasible cascade (kW)	Feasible cascade (kW)	
401.0				0.0	2,483.3	
393.0	8.0	-35.282	-282.3	-282.3	2,201.1	<- QH,min = 2,483.3 kW added
267.0	126.0	-7.045	-887.7	-1,170.0	1,313.4	
262.0	5.0	-13.067	-65.3	-1,235.3	1,248.1	
254.0	8.0	-7.045	-56.4	-1,291.7	1,191.7	
186.0	68.0	14.314	973.3	-318.3	2,165.0	
160.0	26.0	-37.174	-966.5	-1,284.9	1,198.5	
56.0	104.0	-11.525	-1,198.5	-2,483.3	0.0	<- PINCH
42.0	14.0	23.756	332.9	-2,150.5	332.9	
34.0	8.0	75.244	602.0	-1,548.5	934.9	<- QC,min = 934.9 kW rejected

PTA summary

Min. hot utility QH,min	2,483.3 kW
Min. cold utility QC,min	934.9 kW
Pinch (shifted)	56.0 °C
Hot pinch temperature	60.0 °C
Cold pinch temperature	52.0 °C

Shifted stream temperatures, the full heat cascade with the pinch row highlighted, and the PTA summary.

Column	Meaning
T* (°C)	Shifted interval boundary temperatures (hot streams $-\Delta T_{min}/2$, cold streams $+\Delta T_{min}/2$), descending.
ΔT , $\Sigma CPh-\Sigma CPc$, ΔH	Interval width, net heat-capacity flowrate of the streams present, and interval heat surplus/deficit.
Infeasible cascade	Cumulative surplus with zero hot utility — the most negative value fixes QH,min.
Feasible cascade	Cascade after adding QH,min at the top; it is zero at the pinch and equals QC,min at the bottom.

4.2 Results — full target set (live)

HEN Targeting Results

$\Delta T_{min} = 8.0\text{ }^{\circ}\text{C}$ | SnapTarget (TOP v9 model) | live: updates with the Streams sheet

Pinch & Energy Targets

Quantity	Value	Units
ΔT_{min}	8.0	$^{\circ}\text{C}$
Pinch (shifted) temperature	56.0	$^{\circ}\text{C}$
Hot pinch temperature	60.0	$^{\circ}\text{C}$
Cold pinch temperature	52.0	$^{\circ}\text{C}$
Min. hot utility QH,min	2,483.3	kW
Min. cold utility QC,min	934.9	kW
Heat recovery Qrec	17,132.7	kW
Total hot duty Qhot,total	18,067.6	kW

Area & Units Targets

Quantity	Value	Units
Area target (counter-current)	2,863.1	m^2
Area (Ft-corrected)	3,210.8	m^2
Min. units target Nu	10	-
Number of shells Nsh	21.3	-
LMTD correction Ft	0.892	-

Cost Targets

Quantity	Value	Units
OPEX (utilities)	\$325,118	\$/yr
Ann. CAPEX (targeting)	\$36,192	\$/yr
Ann. CAPEX (design-adj.)	\$70,202	\$/yr
TAC (targeting)	\$361,309	\$/yr
TAC (design-adjusted)	\$395,320	\$/yr
Capital recovery factor (CRF)	0.16275	-

Three sections — Pinch & Energy Targets, Area & Units Targets, Cost Targets. Every quantity is a live ST_TARGET formula on the ΔT_{min} cell — the demo case at $\Delta T_{min} = 8\text{ }^{\circ}\text{C}$ is shown.

4.3 Supertargeting Trade-Off Study (live)

Supertargeting Trade-Off Study (0.5-40 °C)

Targeting optimum = 0.5 °C (TAC \$295,333/yr) Design-adjusted optimum = 8.0 °C (TAC \$395,320/yr)

Live: the whole table, the optima and the charts recalculate when the Streams sheet changes.

ΔTmin °C	QH,min kW	QC,min kW	Qrec kW	Area m²	Nu -	OPEX \$/yr	ACAPEX (t) \$/yr	TAC (t) \$/yr	Nsh -	Ft -	Area (Ft) m²	ACAPEX (d) \$/yr	TAC (d) \$/yr
0.50	1,919.0	370.5	17,697.0	8,898.4	10	\$248,975	\$46,358	\$295,333	592.3	0.300	29,661.2	\$1,769,001	\$2,017,975
0.75	1,937.8	389.3	17,678.2	7,751.0	10	\$251,513	\$44,685	\$296,198	364.1	0.337	23,015.1	\$1,105,346	\$1,356,859
1.00	1,956.6	408.1	17,659.4	7,004.0	10	\$254,051	\$43,549	\$297,600	257.8	0.377	18,580.3	\$790,676	\$1,044,727
1.50	1,994.3	445.8	17,621.8	6,032.8	10	\$259,127	\$42,006	\$301,133	158.5	0.450	13,401.3	\$492,661	\$751,788
2.00	2,031.9	483.4	17,584.2	5,395.4	10	\$264,203	\$40,946	\$305,149	112.2	0.515	10,481.0	\$352,000	\$616,203
2.50	2,069.5	521.0	17,546.6	4,928.1	10	\$269,280	\$40,140	\$309,420	85.9	0.572	8,618.7	\$271,165	\$540,444
3.00	2,107.1	558.6	17,508.9	4,563.2	10	\$274,356	\$39,493	\$313,849	69.0	0.622	7,335.1	\$219,120	\$493,476
3.50	2,144.8	596.3	17,471.3	4,266.5	10	\$279,432	\$38,953	\$318,385	57.3	0.667	6,401.2	\$183,020	\$462,452
4.00	2,182.4	633.9	17,433.7	4,018.1	10	\$284,508	\$38,490	\$322,999	48.8	0.706	5,693.8	\$156,625	\$441,133
4.50	2,220.0	671.5	17,396.1	3,805.8	10	\$289,584	\$38,087	\$327,672	42.4	0.740	5,141.0	\$136,550	\$426,135
5.00	2,257.6	709.1	17,358.4	3,621.3	10	\$294,661	\$37,730	\$332,391	37.4	0.771	4,698.1	\$120,809	\$415,470
6.00	2,332.9	784.4	17,283.2	3,314.4	10	\$304,813	\$37,122	\$341,935	30.0	0.821	4,034.5	\$97,795	\$402,607
7.00	2,408.1	859.6	17,208.0	3,067.4	10	\$314,965	\$36,619	\$351,584	25.0	0.861	3,562.7	\$81,851	\$396,816
8.00	2,483.3	934.9	17,132.7	2,863.1	10	\$325,118	\$36,192	\$361,309	21.3	0.892	3,210.8	\$70,202	\$395,320
9.00	2,558.6	1,010.1	17,057.5	2,690.6	10	\$335,270	\$35,822	\$371,092	18.5	0.916	2,938.4	\$61,345	\$396,615
10.00	2,633.8	1,085.3	16,982.2	2,542.5	10	\$345,422	\$35,498	\$380,921	16.3	0.934	2,721.2	\$54,399	\$399,821
11.00	2,709.1	1,160.6	16,907.0	2,413.6	10	\$355,575	\$35,211	\$390,786	14.5	0.949	2,543.7	\$48,816	\$404,390
12.00	2,784.3	1,235.8	16,831.7	2,300.4	10	\$365,727	\$34,955	\$400,682	13.1	0.960	2,395.9	\$44,236	\$409,963
13.00	2,859.6	1,311.1	16,756.5	2,200.0	10	\$375,880	\$34,723	\$410,603	11.9	0.969	2,270.5	\$40,416	\$416,295
14.00	2,934.8	1,386.3	16,681.2	2,110.3	10	\$386,032	\$34,514	\$420,546	10.9	0.976	2,162.6	\$37,183	\$423,215
15.00	3,010.1	1,461.6	16,606.0	2,029.7	10	\$396,184	\$34,322	\$430,506	10.0	0.981	2,068.6	\$34,415	\$430,599
16.00	3,085.3	1,536.8	16,530.8	1,956.7	10	\$406,337	\$34,147	\$440,483	10.0	0.985	1,985.8	\$34,217	\$440,554
18.00	3,235.8	1,687.3	16,380.3	1,829.8	10	\$426,642	\$33,836	\$460,478	10.0	0.991	1,846.2	\$33,877	\$460,518
20.00	3,386.3	1,837.8	16,229.8	1,723.2	10	\$446,946	\$33,569	\$480,516	10.0	0.995	1,732.6	\$33,593	\$480,539
22.00	3,536.8	1,988.3	16,079.3	1,632.5	10	\$467,251	\$33,338	\$500,589	10.0	0.997	1,637.9	\$33,351	\$500,603
24.00	3,687.3	2,138.8	15,928.8	1,554.4	10	\$487,556	\$33,134	\$520,690	10.0	0.998	1,557.4	\$33,143	\$520,698
26.00	3,837.7	2,289.2	15,778.3	1,486.4	10	\$507,861	\$32,955	\$540,815	10.0	0.999	1,488.2	\$32,960	\$540,820
28.00	3,988.2	2,439.7	15,627.8	1,426.8	10	\$528,165	\$32,795	\$560,960	10.0	0.999	1,427.9	\$32,798	\$560,963
30.00	4,138.7	2,590.2	15,477.3	1,374.2	10	\$548,470	\$32,652	\$581,122	10.0	1.000	1,374.8	\$32,654	\$581,124
32.00	4,289.2	2,740.7	15,326.9	1,327.5	10	\$568,775	\$32,523	\$601,298	10.0	1.000	1,327.8	\$32,524	\$601,299
34.00	4,439.7	2,891.2	15,176.4	1,285.7	10	\$589,080	\$32,407	\$621,486	10.0	1.000	1,285.9	\$32,407	\$621,487
36.00	4,590.2	3,041.7	15,025.9	1,248.2	10	\$609,384	\$32,301	\$641,686	10.0	1.000	1,248.3	\$32,301	\$641,686
38.00	4,740.7	3,192.2	14,875.4	1,214.5	10	\$629,689	\$32,205	\$661,894	10.0	1.000	1,214.5	\$32,205	\$661,894
40.00	4,891.2	3,342.7	14,724.9	1,184.4	10	\$649,994	\$32,119	\$682,113	10.0	1.000	1,184.4	\$32,119	\$682,113

<- targeting optimum

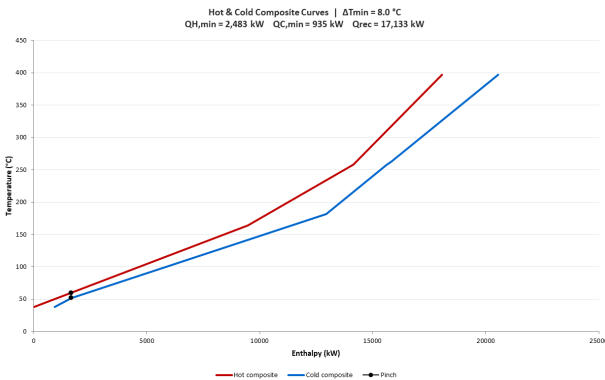
<- design-adjusted optimum

ΔTmin sweep 0.5–40 °C. Green row = targeting-level optimum (min TAC with pure counter-current area); blue row = design-adjusted optimum (min TAC with Ft + shells correction). The highlighting is conditional formatting, so it moves automatically when the inputs change.

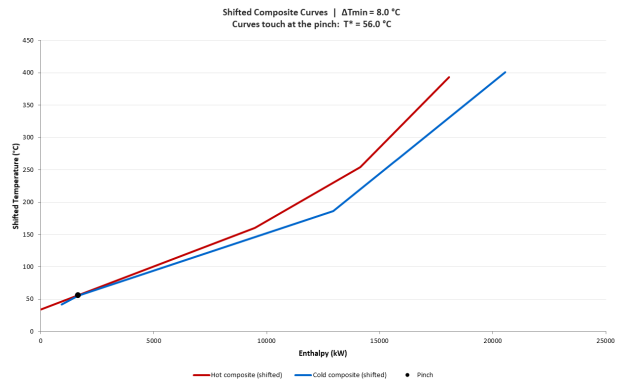
Column group	Meaning
QH,min / QC,min / Qrec	PTA energy targets and process-to-process heat recovery at each ΔTmin.
Area / Nu	Bath-formula counter-current area target and minimum number of units (Euler, MER).
OPEX / ACAPEX (t) / TAC (t)	Utility cost, annualized capital and total annualized cost of the targeting-level model.
Nsh / Ft / Area (Ft)	Estimated 1–2 shells, average LMTD correction factor, Ft-corrected area.
ACAPEX (d) / TAC (d)	Annualized capital and TAC of the design-adjusted model (shells + Ft).

4.4 Charts (live, one per sheet)

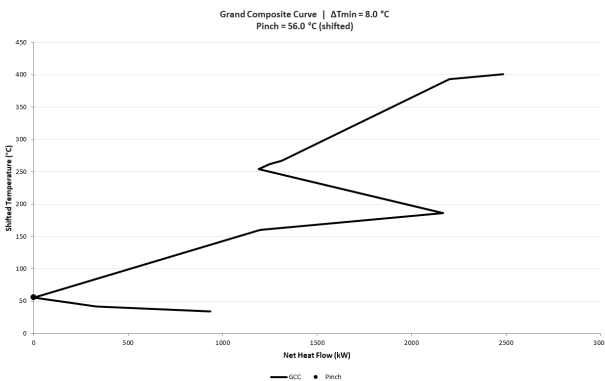
Each chart lives on its own sheet — Composite Curves, Shifted Composites, Grand Composite, Trade-Off (Targeting), Trade-Off (Design), Energy Targets, Area Targets — so the workbook stays organized. All seven read their data and captions from formulas: titles show the current QH/QC/Qrec, pinch and optima and follow every input change.



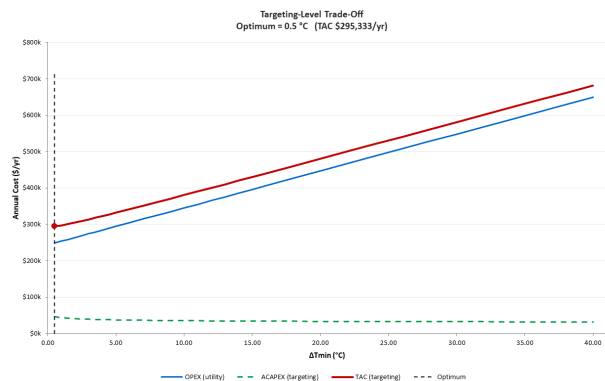
Composite curves with the pinch gap marked



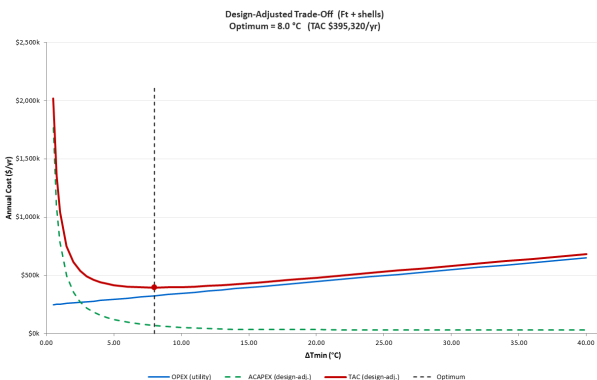
Shifted composite curves touching at the pinch



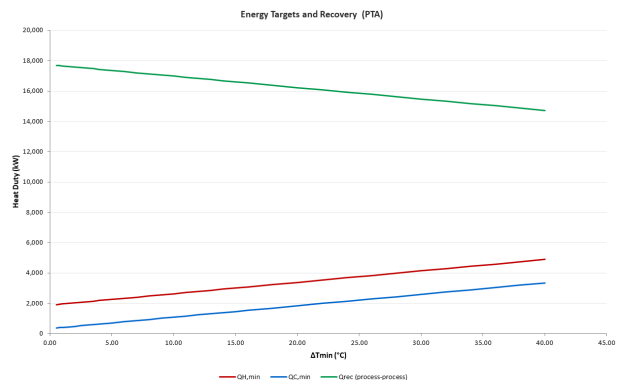
Grand composite curve with the pinch point



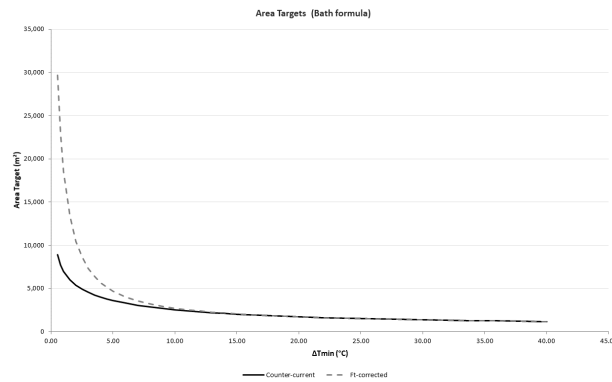
Targeting-level trade-off with its optimum



Design-adjusted trade-off with its optimum



Energy targets vs ΔT_{min}



Area targets vs ΔT_{min}

5. Worksheet function reference

All functions read the stream table and economics from the Streams sheet of the workbook they are used in, and recalculate automatically (volatile). They work in any workbook created by New Template.

Function	Returns	Example
ST_TARGET(dtmin, key)	Any single target at the given ΔT_{min} — see the key table below.	=ST_TARGET(8, "tac_d") → 395,320
ST_QHMIN(dtmin)	Minimum hot utility (kW).	=ST_QHMIN(8) → 2,483.3
ST_QCMIN(dtmin)	Minimum cold utility (kW).	=ST_QCMIN(8) → 934.9
ST_PINCH(dtmin)	Pinch temperature, shifted scale (°C).	=ST_PINCH(8) → 56.0
ST_AREA(dtmin)	Counter-current area target (m²).	=ST_AREA(8) → 2,863.1
ST_UNITS(dtmin)	Minimum number of units (—).	=ST_UNITS(8) → 10
ST_TAC(dtmin, [model])	Total annualized cost; model "d" (default, design-adjusted) or "t" (targeting).	=ST_TAC(8, "t") → 361,309
ST_OPT(model, what)	Optimum of the 0.5–40 °C sweep; model "t"/"d"; what "dt", "tac" or "tacmax".	=ST_OPT("d", "dt") → 8.0
ST_PINCHH(dtmin)	Hot-composite enthalpy at the hot pinch temperature (kW) — x-position of the pinch marker.	=ST_PINCHH(8)
ST_CURVE(kind, dtmin)	60×2 curve block (X, Y) padded with #N/A; kind "hot", "coldshift", "gcc", "hot_s" or "cold_s" (shifted composites). Enter over a 60-row × 2-column range.	=ST_CURVE("gcc", 8)

Keys accepted by ST_TARGET

Key	Quantity	Units	Key	Quantity	Units
qh	Min. hot utility QH,min	kW	opex	Utility operating cost	\$/yr
qc	Min. cold utility QC,min	kW	acapex_t	Ann. capital, targeting	\$/yr
qrec	Heat recovery (process–process)	kW	acapex_d	Ann. capital, design-adj.	\$/yr
qhot_total	Total hot-stream duty	kW	tac_t	TAC, targeting	\$/yr
area	Area target (counter-current)	m²	tac_d	TAC, design-adjusted	\$/yr
area_ft	Area, Ft-corrected	m²	pinch	Pinch (shifted)	°C
nu	Min. units (Euler MER)	—	pinch_hot	Hot pinch temperature	°C
nsh	Number of 1–2 shells	—	pinch_cold	Cold pinch temperature	°C
ft	LMTD correction factor	—	crf	Capital recovery factor	—

6. Reading the results — demo case

The two cost models bracket reality: the targeting-level model assumes pure counter-current exchangers at the minimum unit count, so its optimum sits at a very small $\Delta T_{\min}$; the design-adjusted model adds the F_t correction and a shells estimate, which penalizes small approach temperatures and moves the optimum to a practical value — in agreement with the Aspen Energy Analyzer result ($\approx 9^\circ\text{C}$) for this plant.

Quantity (kerosene hydrotreater demo)	Value
Targeting-level optimum	$\Delta T_{\min} = 0.5^\circ\text{C}$, TAC = \$295,333/yr
Design-adjusted optimum	$\Delta T_{\min} = 8.0^\circ\text{C}$, TAC = \$395,320/yr
At $\Delta T_{\min} = 8^\circ\text{C}$	QH,min = 2,483.3 kW, QC,min = 934.9 kW, Qrec = 17,132.7 kW
Pinch at 8°C	56.0 $^\circ\text{C}$ shifted (hot 60.0 $^\circ\text{C}$ / cold 52.0 $^\circ\text{C}$)
Area / units at 8°C	2,863.1 m ² counter-current (3,210.8 m ² Ft-corrected), Nu = 10, Nsh = 21.3, $F_t = 0.892$
Costs at 8°C	OPEX \$325,118/yr, ACAPEX \$36,192 (t) / \$70,202 (d), CRF = 0.16275

7. Model and validation

Energy targets: Problem Table Algorithm on shifted temperatures. Area target: Bath formula (vertical heat-transfer model) on balanced composite curves with the Chen LMTD approximation and stream-individual film coefficients. Units target: Euler ($N-1$) with MER pinch decomposition. Design adjustment: shells power-law (reference 15°C , exponent 1.2) and $F_t = 1 - 0.8 \cdot \exp(-0.25 \cdot \Delta T_{\min})$ floored at 0.30 (Ahmad–Linnhoff–Smith supertargeting). Economics: CAPEX = $N \cdot (16,506 + 145 \cdot A_{\text{per}}^{0.65})$ \$, annualized with CRF at 10 % over 10 years; utilities 128.5 / 6.426 \$/kW·yr.

The add-in is a line-by-line port of the TOP v9 Python model (trade_off_plot_9.py), whose energy and area targets are validated against OpenPinch (energy within 0.04 %, area within 6 %). The VBA engine reproduces the full Python sweep — 34 $\Delta T_{\min}$ points $\times$ 14 quantities — to a worst relative deviation of 4×10^{-15} (machine precision).

SnapTarget v2.5 — created by Mohammed Ali Mansoury, July 2026. Built from the TOP v9 supertargeting model of the Kerosene Hydrotreater HEN graduation project.